

Unit 11, Lesson 2: Subtracting Rational Numbers



Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

☐ I can fluently subtract rational numbers.



11.2.1 Warm-Up: Math Talk

Several students discussed which fraction was greater, $\frac{13}{19}$ or $\frac{2}{3}$. A summary of the discussion is shown.

Amy:	I changed $\frac{13}{19}$ to $\frac{13}{18}$, and since $\frac{2}{3}$ is equal to $\frac{12}{18}$, I'm about 70% confident that $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Brandon:	The fractions were small enough that I could do mental math and change them into common denominators, so $\frac{13}{19} = \frac{39}{57}$, and $\frac{2}{3} = \frac{38}{57}$. Therefore, $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Carlos:	I changed $\frac{13}{19}$ to $\frac{14}{20}$. Then I thought of a pizza, and that's a lot of little pieces. So, I'm about 70% confident that $\frac{2}{3}$ is bigger than $\frac{13}{19}$.
Dimitri:	I know $\frac{2}{3}$ is $\frac{12}{18}$ if I multiply the top and bottom by 6. $\frac{12}{18}$ is bigger than $\frac{13}{19}$. I'm 90% confident.

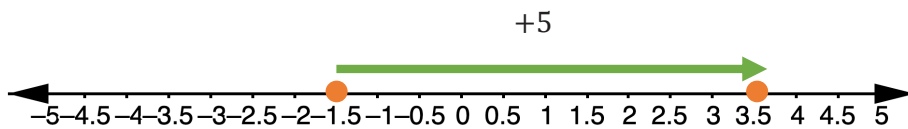
1. Explain which student's reasoning you agree with.
2. Choose one student you didn't agree with, and explain why you did not agree.



11.2.2 Guided Instruction: Understanding Differences

Finding the **difference** of two numbers means to *subtract the numbers*. When finding the difference between two numbers, subtract the numbers in the order they are named. For example, the difference of 3.5 and -1.5 is represented by the subtraction expression $3.5 - (-1.5)$.

The difference of two numbers is how far apart the numbers are on a number line.



1. The numbers 3.5 and -1.5 are _____ units apart. The direction to move from the second number (-1.5) to the first number (3.5) is to the ☐ left, ☐ right, or the ☐ positive ☐ negative direction. Therefore $3.5 - (-1.5) = +5$.

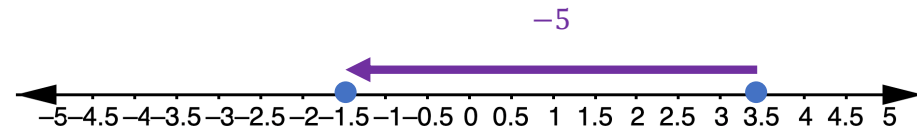
2. Suppose the numbers are subtracted in the opposite order, $(-1.5) - 3.5$.

The numbers 3.5 and -1.5 are still _____ units apart, but the direction of the move has changed. To move from

the second number (3.5) to the first number (-1.5) is to

move to the ☐ left ☐ right or the ☐ positive ☐ negative direction.

Therefore, $(-1.5) - 3.5 = -5$.





11.2.3 Collaboration: Understanding Differences

- Working with your partner, write each phrase as a subtraction expression. Then, draw a number line representation to find the value of the expression. The first phrase has been done for you.

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{1}{5}$ and $-\frac{4}{5}$	$-\frac{1}{5} - (-\frac{4}{5})$	
Value = $\frac{3}{5}$		

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{4}{5}$ and $-\frac{1}{5}$		
Value = _____		

Phrase	Subtraction Expression	Number Line Representation
The difference of 0.9 and -0.5		
Value = _____		

Phrase	Subtraction Expression	Number Line Representation
The difference of -0.5 and 0.9		
Value = _____		

2. With your partner, complete the table to find the difference of the values.

Expression	Distance Between the Values	Direction from Second Value to First Value	Difference
$9.9 - 6.2$	3.7	Right/ positive	$9.9 - 6.2 = 3.7$
$9.9 - (-6.2)$			$9.9 - (-6.2) = \underline{\hspace{2cm}}$
$(-9.9) - 6.2$			$(-9.9) - 6.2 = \underline{\hspace{2cm}}$
$(-9.9) - (-6.2)$			$(-9.9) - (-6.2) = \underline{\hspace{2cm}}$
$6.2 - 9.9$			$6.2 - 9.9 = \underline{\hspace{2cm}}$
$6.2 - (-9.9)$			$6.2 - (-9.9) = \underline{\hspace{2cm}}$
$(-6.2) - 9.9$			$(-6.2) - 9.9 = \underline{\hspace{2cm}}$
$(-6.2) - (-9.9)$			$(-6.2) - (-9.9) = \underline{\hspace{2cm}}$



11.2.4 Guided Instruction: Finding Differences

Adding a negative value has the same result as subtracting a positive value. This is because addition and subtraction have an inverse relationship.

For example, $5 + (-6)$ is equivalent to $5 - 6$. Both expressions have a value of -1 .

This relationship can be used to subtract rational numbers. The difference of two numbers can be rewritten as an equivalent addition expression. Then, the rules for addition of two numbers can be followed.

1. Fill in the blanks to complete the table.

Original Expression	$1\frac{2}{5} - (-\frac{6}{7})$
Rewrite as an equivalent addition problem by keeping the first term, changing subtraction to addition, and changing the second term to its opposite.	$1\frac{2}{5} \square (\square \frac{6}{7})$
Solve the problem using the rules of addition.	$\square$



11.2.5 Collaboration: Does the Order Matter?

- With your partner, determine who will be Partner A and who will be Partner B. Partner A will work the problems in Column A while Partner B works the problems in Column B. Then, work together to answer the questions that follow.

Column A	Value	Column B	Value
$3 - 2$		$2 - 3$	
$5 - (-9)$		$(-9) - 5$	
$(-11) - 2$		$2 - (-11)$	
$(-\frac{1}{6}) - (-\frac{1}{3})$		$(-\frac{1}{3}) - (-\frac{1}{6})$	
$(-1.2) - (-3.6)$		$(-3.6) - (-1.2)$	
$(-2\frac{1}{2}) - (-3\frac{1}{2})$		$(-3\frac{1}{2}) - (-2\frac{1}{2})$	

- Explain the difference between the expressions in Column A and the expressions in Column B.
- Explain the relationship between the values of the expressions in Column A and the value of the expressions in Column B.



11.2.6 Your Turn!

- What is the difference of $\frac{9}{14}$ and $-\frac{13}{20}$?
- Find the difference for each expression.
 - $5\frac{7}{8} - (-2\frac{2}{5})$
 - $-45.6 - (-1.772)$



11.2.7 Wrap-Up: Growth Mindset

Making mistakes is one of the best ways to learn. Reflecting on our mistakes helps our brains to look for that mistake again and to avoid it the next time.

1. What mistake did you make today that you can learn from?

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

Unit 11, Lesson 3: Multiplying Rational Numbers



Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can fluently multiply rational numbers.